

Contrastive Learning and Zero-Shot Transfer

DATA 443/643 · Week 4, note 4.3

Week 4

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Ideas from Stanford CS231N 2025 L16 (InfoNCE, zero-shot prompts). Original slides; that course is not copied.

Lecture plan

- 1 An N by N grid; the diagonal is the match
- 2 InfoNCE; the rest of the batch are negatives
- 3 Zero-shot: nearest prompt, not a C-way softmax
- 4 Linear probe versus zero-shot

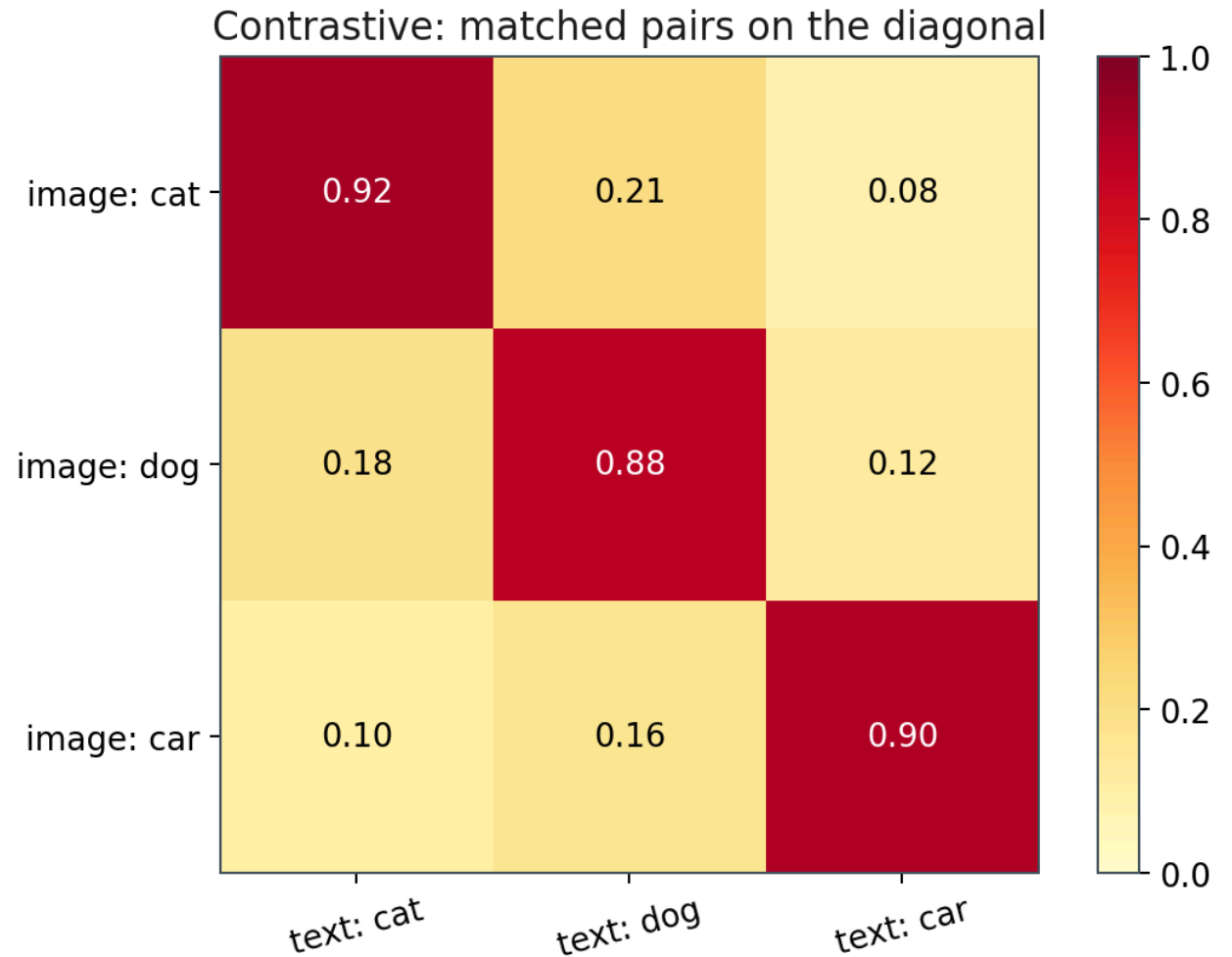
First time: what / why / formula

- What / why. InfoNCE on a batch of N matched pairs. Zero-shot: embed class phrases, pick $\arg \max_c \cos(f(x), g(\text{prompt}_c))$. You may not have labels for every class at train time. A similarity space transfers to new names.
- Architecture. Two encoders (or two views of one). A batch matrix of scores. Softmax over the row (or bidirectional).
- Formula. $\mathcal{L} = -\frac{1}{N} \sum_i \log \frac{\exp(s_{ii} / \tau)}{\sum_j \exp(s_{ij} / \tau)}$, $s_{ij} = \cos(u_i, v_j)$.
- Tradeoffs. + No C -way head, works with prompts. – Needs big batches (many negatives); temperature τ is a knob; compositionality is not free.

How. Lab 4: four pairs, heatmap of the $N \times N$ scores. The diagonal should win.

Matched pairs on the diagonal

- Encode N images and N captions.
- Image i should sit with caption i.
- This is 4.1's coordinated space, with a loss.



InfoNCE, one row

$$-\log \frac{\exp(\text{sim}(v_i, t_i) / \tau)}{\sum_j \exp(\text{sim}(v_i, t_j) / \tau)}$$

CLIP trains this both ways: image-to-text and text-to-image.

Small tau sharpens. Argmax of a row does not flip.

N=4 in lab is to see the matrix, not to match ImageNet.

A 2 by 2 you will write

- Scores $[0.9, 0.1]$, $\tau = 1$, gives about $[0.69, 0.31]$.
- Loss is $-\log 0.69$ about 0.37.
- Identical captions: uniform softmax, no learning.

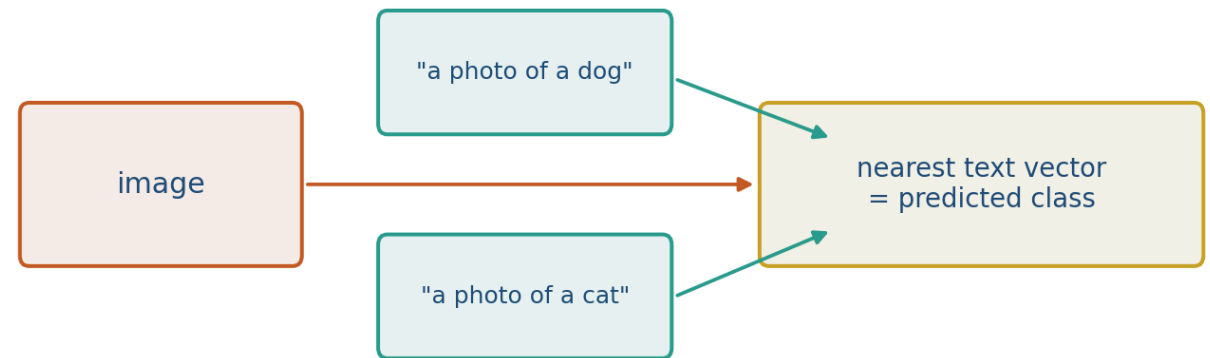
InfoNCE: the rest of the batch are negatives. The diagonal must be special.

row 0 scores	softmax	$-\log p_{\text{diag}}$
$[0.9, 0.1], \tau = 1$	$[0.69, 0.31]$	0.37
identical captions	uniform	cannot learn

Zero-shot is nearest text

- Encode "a photo of a dog", not the token dog.
- New classes are new strings.
- Unseen phrases do not magically work.

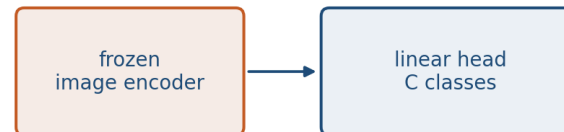
Zero-shot: class names are text embeddings, not trained heads



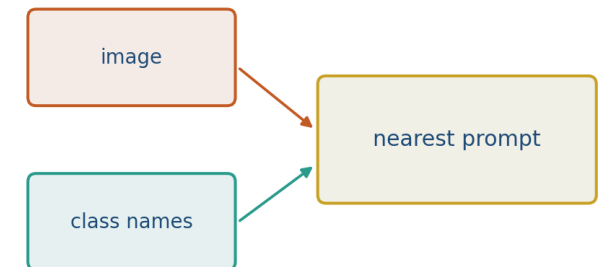
Two ways to transfer

- Linear probe: freeze the image tower, train C classes.
- Zero-shot: the classifier is the text tower.
- Captioning and LLaVA wait. This week is the score.

Linear probe: train a softmax



Zero-shot: no trained C-way head



What to take from this lecture

Off-diagonal is a negative

Not a don't-care. Shuffle in Lab 4 and the loss should rise.

Prompt it

A photo of beats the raw label.
Ensemble is optional.

Week 5

CLIP's data, retrieval, and the bias that comes with the geometry.